Salahaddin University-Erbil
College of Computer Science and IT
Stage 2 Cs & IT

Web Technology Basics - Lec. 3 – HTML Forms

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HTML Forms

- Forms add the ability to web pages to not only provide the person viewing the document with dynamic information, but also to obtain information from the person viewing it, and process the information.
- A form is basically a Web page with **input** fields that allows users to enter information.
- When the form is submitted, that information is packaged up and sent off to a Web server to be processed by a Web application. When information processing is done, user will get another Web page.
- An HTML form can contain **input elements** like **text fields**, **checkboxes**, **radio-buttons**, **submit buttons** and more.
- A form can also contain select **lists**, **textarea**, **fieldset**, **legend**, and **label elements**.
- The <**form**> tag is used to create an HTML form.

HTML Form Attributes

<**form** **action**="/action_page.php“ **method**=“post” **target**=“_blank”>

1. **The Action Attribute :** Is the **URL** of the program (CGI) that is going to accept the data from the form, process it, and send a response back to the browser.

2. **Method : GET** (default) or **POST** specifies which **HTTP** method will be used to send the form’s contents to the web server. The CGI application should be written to accept the data from either method.
3. **target :** is the target frame where the response page will show up.

_blank Opens the linked document in a new window or tab

_self Opens the linked document in the same frame as it was clicked (this is default)

HTML <Form> Elements

- The HTML <form> element can contain one or more of the following form elements:
  - **<input>**
  - **<label>**
  - <select>
  - **<textarea>**
  - <button>
  - <fieldset>
  - <legend>
  - <datalist>
  - <output>
  - <option>
  - <optgroup>

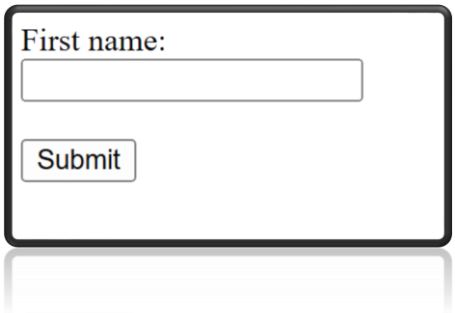
Syntax: <form>
input elements
.
.
</form>

The <input> Element

- The most important form element is the <input> element.
- The <input> element is used to select user information.
- An <input> element can be viewed in many ways, depending on the type attribute.
- An <input> element can be of **type** text field, checkbox, password, radio button, submit button, and more.

- **Example**:

```
<form action="/action_page.php">
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname"><br><br>
  <input type="submit" value="Submit">
</form>
```



Input Element Attributes

1. **Type**: Specifies the type of <input> element to display. The default type is: text.
   - It could take values "text, password, checkbox, radio, button, …etc."
   - **Example:**

```
<form >
    Username: <input type="text" name="usrname"><br>
</form>
```

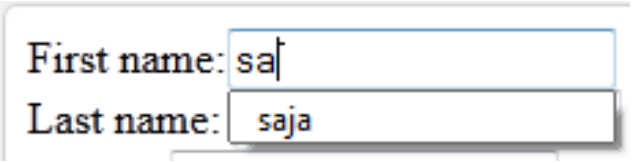
2. **Autocomplete**: specifies whether or not an input field should have autocomplete enabled. Autocomplete allows the browser to predict the value. When a user starts to type in a field, the browser should display options to fill in the field, based on earlier typed values.

Input Element Attributes

Note: The autocomplete attribute works with the following <input> types: text, search, url, tel, email, password, datepickers, range, and color.

Autocomplete Example:

```
<form action="lab1.html" method="_post" >
        <label for="fname">First name:</label>
        <input type="text" id="fname" name="fname" autocomplete="on"> <br>
</form>
```



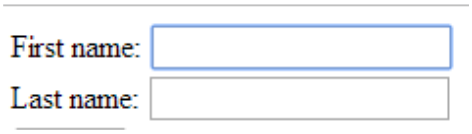
Input Element Attributes

3. **Autofocus**: Specifies that an <input> element should automatically get focus when the page loads.

Example:

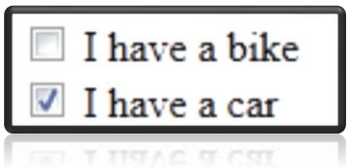
First name: <input type="text" name="fname" **autofocus**>

Last name: <input type="text" name="lname">



4. **Checked**: The checked attribute is a boolean attribute. When present, it specifies that an <input> element should be pre-selected (checked) when the page loads. The checked attribute can be used with <input type="checkbox"> and <input type="radio">.

Input Element Attributes



Example:

```
<form >
    <input type="checkbox" name="vehicle" value="Bike"> I have a bike<br>
    <input type="checkbox" name="vehicle" value="Car" checked> I have a car<br>
</form>
```

5. **Disabled:** The disabled attribute is a boolean attribute. When present, it specifies that the <input> element should be disabled. A disabled input element is unusable and un-clickable.
   - The disabled attribute can be set to keep a user from using the <input> element until some other condition has been met (like selecting a checkbox, etc.). Then, a JavaScript could remove the disabled value, and make the <input> element usable.

Input Element Attributes

Tip: Disabled <input> elements in a form will not be submitted.

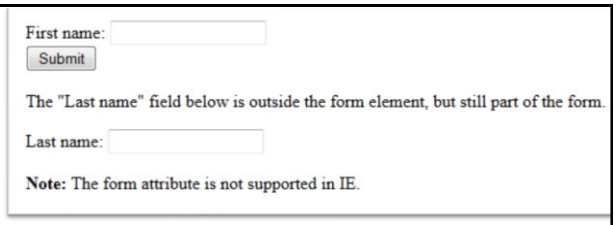
Note: The disabled attribute will not work with <input type="hidden">.

Example:

```
<form >
    First name: <input type="text" name="fname"><br>
    Last name: <input type="text" name="lname" disabled><br>
</form>
```

6. **Form:** An input field located outside the HTML form (but still a part of the form).

Input Element Attributes



Example:

```
<body>
   <form action="demo_form.asp" id="form1">
    <label for="fname">First name:</label>
        <input type="text" id="fname" name="fname“> <br><br>
         <input type="submit" value="Submit">
   </form>

  <p>The "Last name" field below is outside the form element, but still part of the   form.</p>

Last name: <input type="text" name="lname" form="form1">
</body>
```


Input Element Attributes

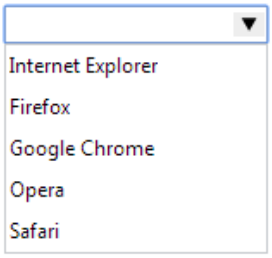
7. **Formtarget**: Specifies where to display the response that is received after submitting the form (for type="submit" and type="image")
   - It may take values like” _blank, _self, framename,…”

Example: First Name:<input type="submit" formtarget="_blank" >

8. List: Refers to a <datalist> element that contains pre-defined options for an <input> element.

Example:

```
<input list="browsers">
        <datalist id="browsers">
           <option value="Internet Explorer">
           <option value="Firefox">
           <option value="Google Chrome">
           <option value="Opera">
           <option value="Safari">
        </datalist>
```

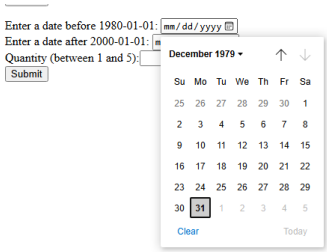


Input Element Attributes

9. **Max and Min:** specifies the maximum and the minimum value for an <input> element.

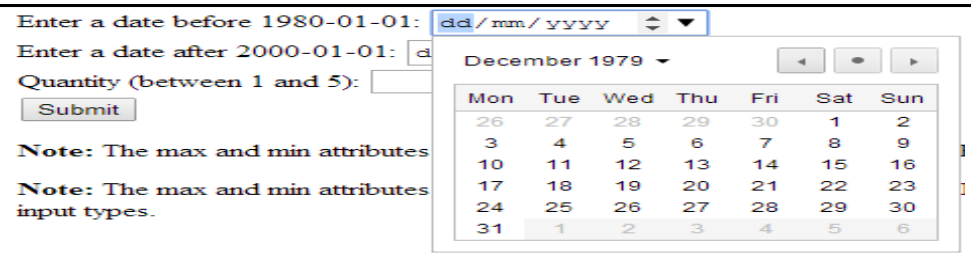
Tip: Use the max attribute together with the min attribute to create a range of legal values.

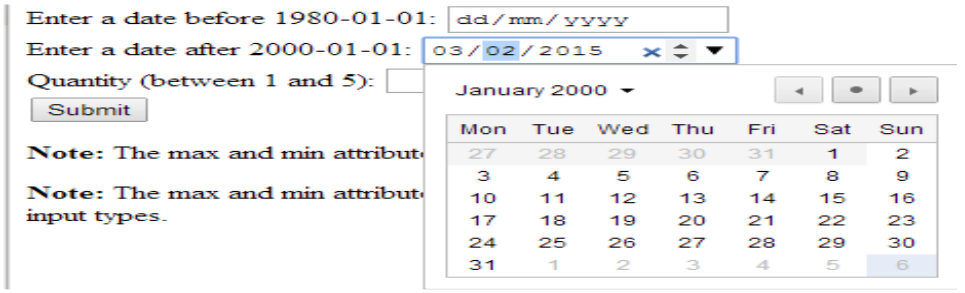
Note1: The max and min attributes works with the following input types: number, range, date, datetime, datetime-local, month, time and week.

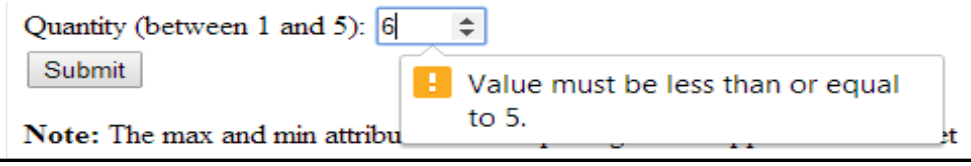


Example:

```
<form action="demo_form.asp">
  Enter a date before 1980-01-01: <input type="date" name="bday" max="1979-12-31"><br>
  Enter a date after 2000-01-01: <input type="date" name="bday" min="2000-01-02"><br>
  Quantity (between 1 and 5):<input type="number“ name="quantity“ min="1" max="5">
<br> <input type="submit">
</form>
```





Input Element Attributes

10.Formaction: The formaction attribute specifies the URL of the file that will process the input control when the form is submitted. The formaction attribute overrides the action attribute of the <form> element.

Note: The formaction attribute is used with type="submit" and type = "image".

Ex: <input type="submit" formaction="demo.asp" value="Submit as admin">

11.Formmethod: The formmethod attribute defines the HTTP method for sending form-data to the action URL.

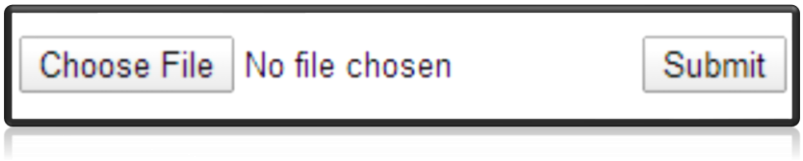
- The formmethod attribute overrides the method attribute of the <form> element. This attribute can be used with type="submit" and type="image".
- The form-data can be sent as URL variables (method="get") or as an HTTP post transaction (method="post").

Input Element Attributes

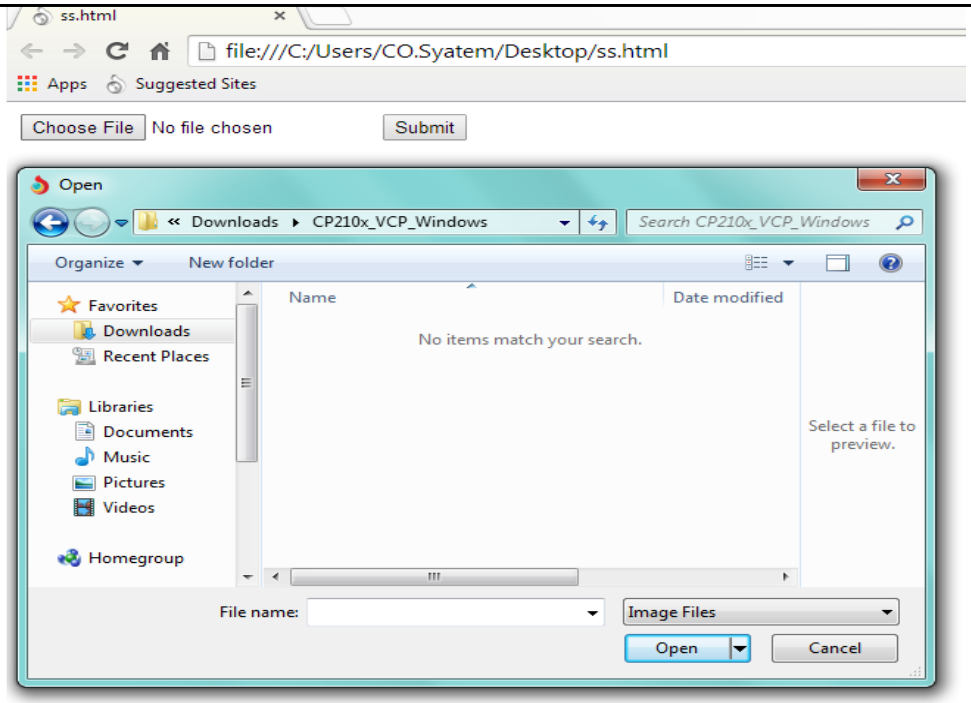
12.Accept: The accept attribute specifies the types of files that the server accepts (that can be submitted through a file upload). The accept attribute can only be used with <input type = "file">.

Example:

```
<form action="demo_form.asp">
  <input type="file" name="pic" accept="image/*">
  <input type="submit">
</form>
```



Note: To specify more than one value, separate the values with a comma (e.g. <input accept = "audio/*,video/*,image/*" />.



Input Element Attributes

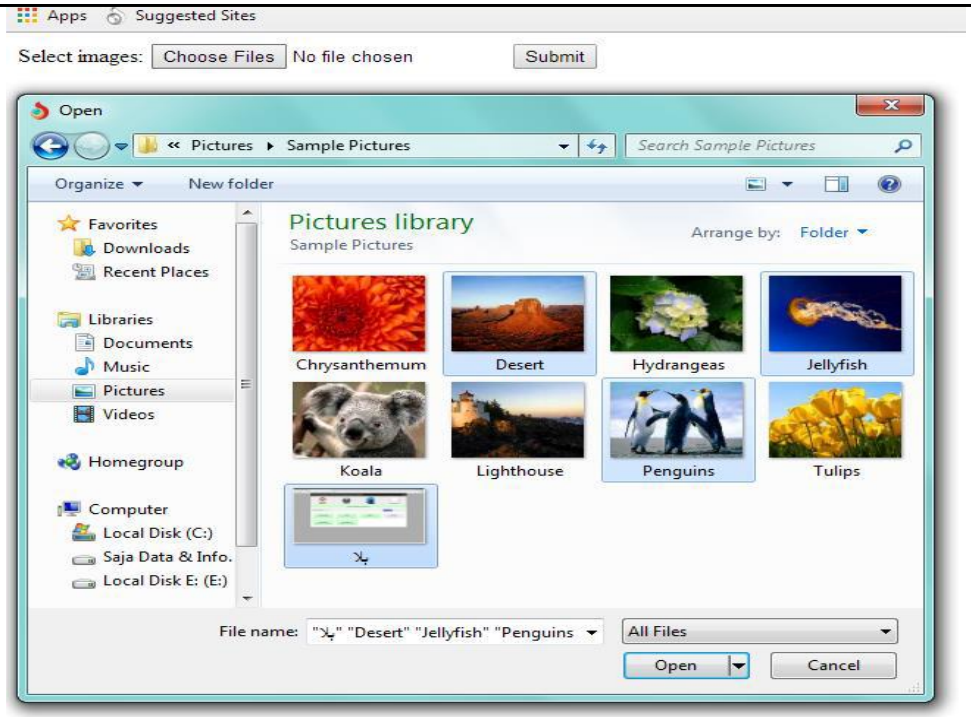
13.Multiple: This attribute is a boolean attribute. When present, it specifies that the user is allowed to enter more than one value in the <input> element. This attribute works with the following input types: email, and file.

Note1: For <input type="file">: to select multiple files, hold down the CTRL or SHIFT key while selecting.

Note2: For <input type="email">: separate each email with a comma, like: mail@example.com, mail2@example.com, mail3@example.com in the email field.

Example:

```
<form action="demo_form.asp">
  Select images: <input type="file" name="img" multiple>
                 <input type="submit">
</form>
```



Input Element Attributes

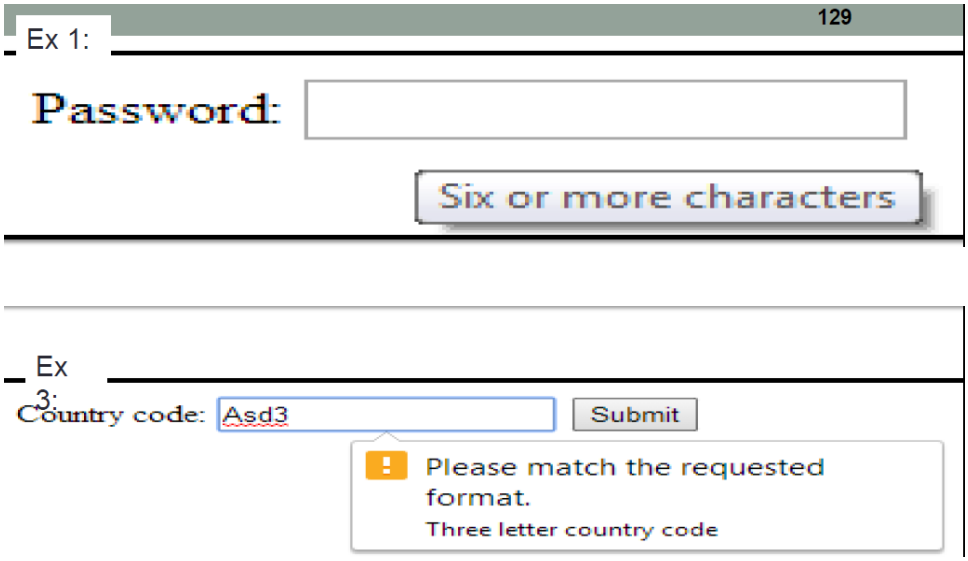
14.Pattern: The pattern attribute specifies a regular expression that the <input> element's value is checked against. This attribute works with the following input types: text, search, url, tel, email, and password.

Example 1: An <input> element with type="password" that must contain 6 or more characters:

Password: <input type="password" name="pw" pattern=".{6,}" title="Six or more characters">

Example 2: Three letter Country code

Country code: <input type="text" name="country_code“ pattern="[A-Za-z]{3}" title="Three letter country code">



Input Element Attributes

15.Value: The value attribute specifies the value of an <input> element. This attribute is used differently for different input types:

- For "**button**", "**reset**", and "**submit**" - it defines the text on the button
- For "**text**", "**password**", and "**hidden**" - it defines the initial (default) value of the input field
- For "**checkbox**", "**radio**", "**image**" - it defines the value associated with the input (this is also the value that is sent on submit)

Note1: The value attribute is required with <input type="checkbox"> and <input type="radio">.

Note2: The value attribute cannot be used with <input type="file">.

Syntax:

<input value="“Text or Number">

Example:

E-mail: E-mail:<input type="email" value= “example@example.com" >

Input Element Attributes

16. Placeholder:

The placeholder attribute specifies a short hint that describes the expected value of an input field (e.g. a sample value or a short description of the expected format). The short hint is displayed in the input field before the user enters a value.

Note: The placeholder attribute works with the following input types: text, search, url, tel, email, and password.

Example:

```
<input type="text" name="fname" placeholder="First name"><br>
<input type="text" name="lname" placeholder="Last name"><br>
```


Input Element Attributes

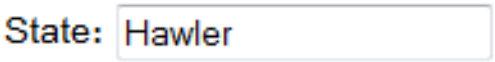
17. Readonly:

The readonly attribute is a boolean attribute. When present, it specifies that an input field is read-only. A read-only input field cannot be modified (however, a user can tab to it, highlight it, and copy the text from it).

The readonly attribute can be set to keep a user from changing the value until some other conditions have been met (like selecting a checkbox, etc.). Then, a JavaScript can remove the readonly value, and make the input field editable.

Syntax:

```
<input readonly>
```



Example:

```
State: <input type="text" name="state" value="Hawler" readonly><br>
```

Input Element Attributes

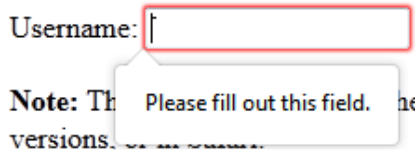
18. Required:

The required attribute is a boolean attribute. When present, it specifies that an input field must be filled out before submitting the form. The required attribute works with the following input types: text, search, url, tel, email, password, date pickers, number, checkbox, radio, and file.

Syntax:

<input required>

Example: Username: <input type="text" name="usrname" required>



The <label> element

- The **<label>** tag defines a label for an **<input>** element.
- The **<label>** element does not render us anything special for the user.
- However, it provides a usability improvement for mouse users, because if the user clicks on the text within the **<label>** element, it toggles the control.
- The for attribute of the **<label>** tag should be equal to the id attribute of the related element to bind them together.

Example:

```
<form action="demo_form.asp">
    <label for="male">Male</label>
    <input type="radio" name="gender" id="male" value="male"><br>
    <label for="female">Female</label>
    <input type="radio" name="gender" id="female" value="female"><br>
    <input type="submit" value="Submit">
</form>
```

- Note: reset button used to set values to default.

HTML <textarea> tag

- The <textarea> tag defines a multi-line text input control.
- A text area can hold an unlimited number of characters, and the text renders in a fixed-width font (usually Courier).
- The size of a text area can be specified by the cols and rows attributes.

❑ **Rows:** This attribute specifies the number of rows of text allowed.

❑ **Cols:** This attribute specifies the number of columns of text allowed.

Example:

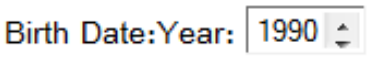
```
<body bgcolor="pink">
    Say Somthing About YourSelf:<br>
    <textarea rows="4" cols="30" name="text">
    about you....
    </textarea>
    <br>
</body>
```

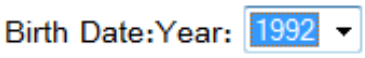
Dropdown list

- **<select>** tag is used to create the list. The whole list content should be enclosed between opened & closed <select>tag.
- **<option>** tag is used to specify the list elements or content.
- The content between the opening <option> and closing </option> tags is what the browsers will display in a drop-down list. However, the value of the **value attribute** is what will be sent to the server when a form is submitted.
- Multiple, size and selected attributes could be used with select.

Example:

```
<form >
    Birth Date:Year: <select size="1" multiple>
    <option value="1990">1990</option>
    <option value="1991">1991</option>
    <option value="1992" selected>1992</option>
    <option value="1993">1993</option>
    <option value="1994">1994</option>
    </select>   </form>
```

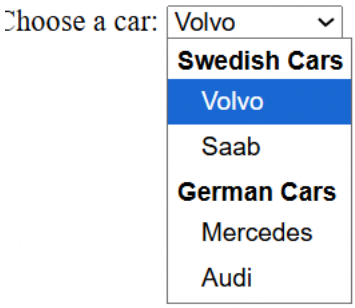




<optgroup>

➢**The <optgroup> tag is used to group related options in a <select> element (drop-down list).**

```
<form action="/action_page.php">
  <label for="cars">Choose a car:</label>
  <select name="cars" id="cars">
    <optgroup label="Swedish Cars">
      <option value="volvo">Volvo</option>
      <option value="saab">Saab</option>
    </optgroup>
    <optgroup label="German Cars">
      <option value="mercedes">Mercedes</option>
      <option value="audi">Audi</option>
    </optgroup>
  </select>
```



Grouping Elements

<**fieldset**> tag is used to group related elements in a form.

<**fieldset**> tag draws a box around the related elements.

<**legend**> tag defines a caption for the <fieldset> element.

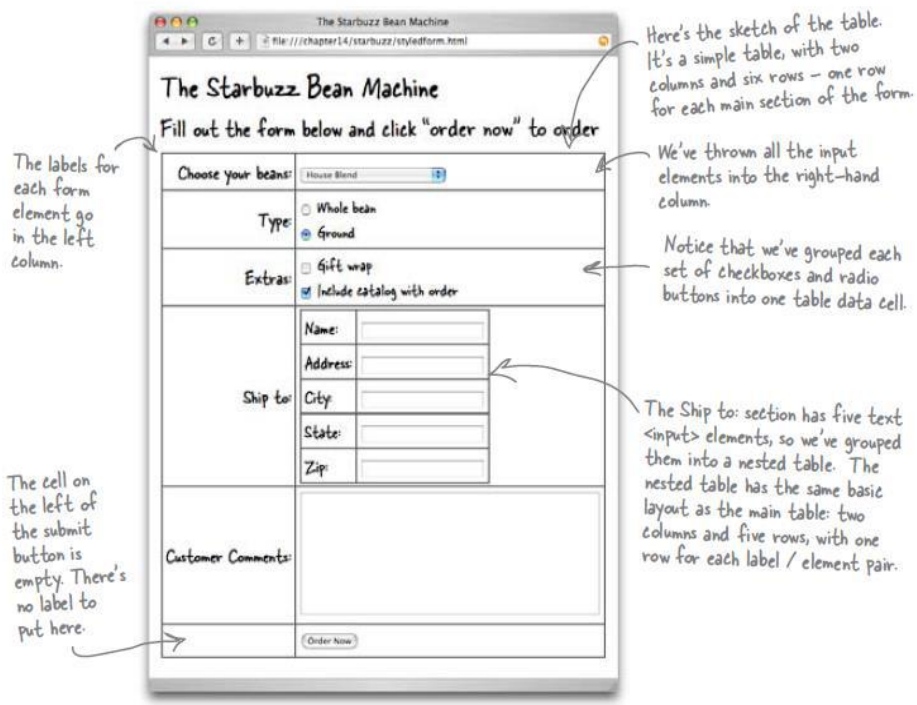
Example:

```
<form action="demo_form.asp">
    <fieldset style="width:200px">
      <legend>Gender:</legend>
      <input type="radio" NAME="GENDER" ID="1" value="1">FEMALE
      <br>
      <input type="radio" NAME="GENDER" ID="0" value="0"> MALE
    </fieldset>
</form>
```

Basic Form Validation (built-in)

- required: field cannot be empty.
- minlength/maxlength: limits text length.
- min/max/step: limits numeric values.
- pattern: a regular expression rule for text.

```
<input name="username" required
minlength="3" maxlength="16">
<input name="age" type="number"
min="16" max="99">
<input name="studentId" pattern="[0-
9]{8}" placeholder="8 digits">
```

- <button type="submit"> sends the form.
- <button type="reset"> clears fields (rarely used).
- <button type="button"> does nothing by default (used with JavaScript).

```
<button type="submit">Create
account</button>
<button type="reset">Clear</button>
<button type="button">Check
availability</button>
```